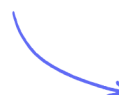


Structured Questions

Ionic & Metallic Bonding & Structure

Evidence of Ions / Ionic Bonding & Structures / Ionic Bond Strength / Polarisation /
Metallic Bonding & Lattices

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Total Marks

/16

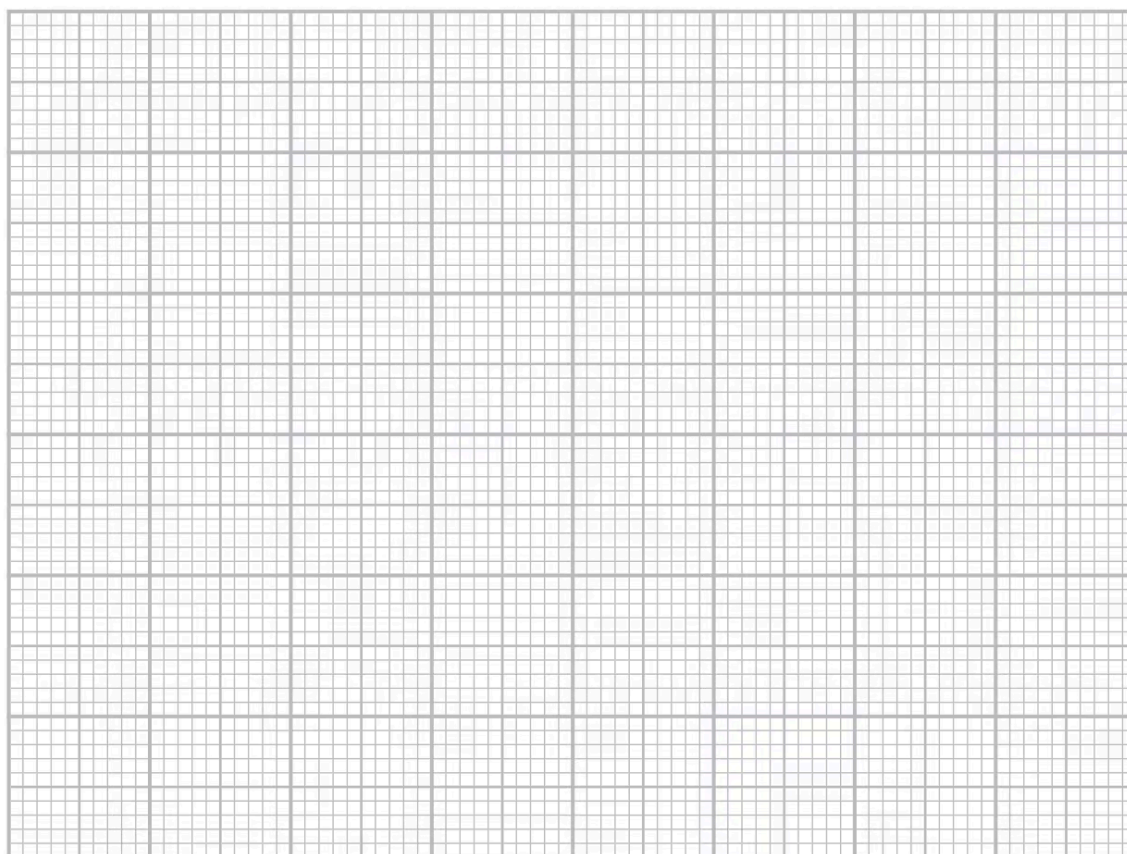
- 1 (a)** This question is about the elements in Period 3 of the Periodic Table, and some of their compounds.

The atomic radii of six of the elements are given.

Symbol	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic number	11	12	13	14	15	16	17	18
Atomic radius / nm	0.191	0.160	0.130			0.102	0.099	0.095

- i) Plot a graph of atomic radius against atomic number.

(2)



ii) Use the graph to estimate the atomic radius of silicon, Si.

(1)

iii) Suggest an explanation for the decrease in atomic radius as atomic number increases across a period.

(3)

(6 marks)

- (b) The melting temperatures of sodium, sodium chloride and chlorine are given in the table.

Complete the table to show the type of structure, the type of bond or force broken on melting and the particles involved.

Substance	Sodium	Sodium chloride	Chlorine
Melting temperature / °C	98	801	-101
Type of structure	giant		simple molecular
Type of bond or force broken on melting			
Particles involved			chlorine molecules

(6 marks)

(c) Solid phosphorus(V) chloride contains PCl_4^+ ions.

i) Draw a dot-and-cross diagram of a PCl_4^+ ion.

Show only outer shell electrons.

(1)

ii) Predict the shape of a PCl_4^+ ion.

Justify your answer.

(3)

Shape

.....

Justification

.....
.....
.....
.....

(4 marks)